# NXTCG Database Administration Guide

**Version 0.0.1**

# Contents

**Chapter 1. Introduction to the NXTCG ECOMAPP Database..... 3**

ECOMAPP Database Architecture.....3

Installing the ECOMAPP database.....5

Adding tables to the ECOMAPP Database.....7

Troubleshooting the NXTCG ECOMAPP database..... 8

**Index.....**

# Chapter 1. Introduction to the NXTCG ECOMAPP Database

The NextGen Trading Card Game (NXTCG) features exclusive collectible cards inspired by the Rocket Software NextGen Academy and its Rocketeers. The platform empowers you to buy and sell new NXTCG cards. Use this guide to install the NXTCG database, add tables, and troubleshoot common issues you may encounter.

ECOMAPP is a Db2 13 for z/OS database for NXTCG. It is compromised of 15 normalized tables with full referential integrity and indexed access paths. Administration is driven by source-controlled DDL and JCL pipelines for build, initialization, statistics, grants, and teardown.

**Key Features:**

- Complete SKU-based product catalog (4,824 trading card variants)
- Multiplier-based dynamic pricing system
- 16 normalized tables with full referential integrity
- Performance indexes on all critical access paths
- Split initialization files for mainframe line length compliance
- Automated JCL for all database operations

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**ECOMAPP Database Architecture**

**Installing the ECOMAPP database**

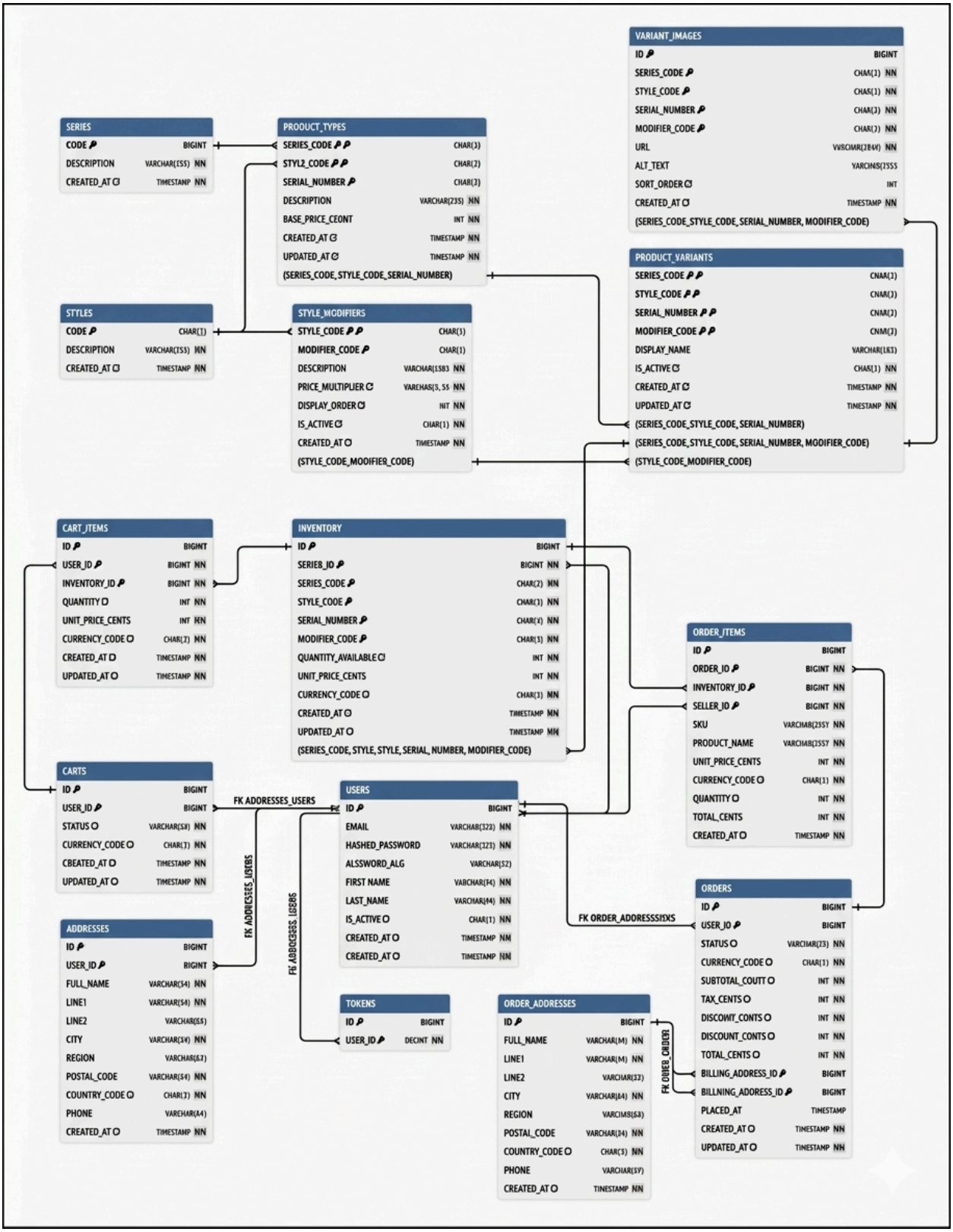
**Adding tables to the ECOMAPP Database**

**Troubleshooting the NXTCG ECOMAPP database**


## ECOMAPP Database Architecture

The database design and architecture includes SKU composition, multiplier-based pricing, and normalized entities with referential integrity. NXTCG's database is hosted on Db2 13 for z/OS.

ECOMAPP is a normalized, SKU-driven data model with centralized pricing logic and strong referential integrity. The catalog supports 4,824 sellable SKUs derived from 104 base product types and 63 style modifiers, designed to be scalable and consistent with updates.


VARIANT_IMAGES
ID BIGINT
SERIES_CODE CHAR(1) NN
STYLE_CODE CHAR(1) NN
SERIAL_NUMBER CHAR(3) NN
MODIFIER_CODE CHAR(3) NN
URL VARCHAR(2048) NN
ALT_TEXT VARCHAR(255)
SORT_ORDER INT
CREATED_AT TIMESTAMP NN
(SERIES_CODE, STYLE_CODE, SERIAL_NUMBER, MODIFIER_CODE)
SERIES
CODE BIGINT
DESCRIPTION VARCHAR(255) NN
CREATED_AT TIMESTAMP NN
PRODUCT_TYPES
SERIES_CODE CHAR(3)
STYLE_CODE CHAR(2)
SERIAL_NUMBER CHAR(3)
DESCRIPTION VARCHAR(255) NN
BASE_PRICE_CENT INT NN
CREATED_AT TIMESTAMP NN
UPDATED_AT TIMESTAMP NN
(SERIES_CODE, STYLE_CODE, SERIAL_NUMBER)
PRODUCT_VARIANTS
SERIES_CODE CHAR(3)
STYLE_CODE CHAR(3)
SERIAL_NUMBER CHAR(3)
MODIFIER_CODE CHAR(3)
DISPLAY_NAME VARCHAR(183)
IS_ACTIVE CHAR(1) NN
CREATED_AT TIMESTAMP NN
UPDATED_AT TIMESTAMP NN
(SERIES_CODE, STYLE_CODE, SERIAL_NUMBER)
(SERIES_CODE, STYLE_CODE, SERIAL_NUMBER, MODIFIER_CODE)
(STYLE_CODE, MODIFIER_CODE)
STYLES
CODE CHAR(1)
DESCRIPTION VARCHAR(255) NN
CREATED_AT TIMESTAMP NN
STYLE_MODIFIERS
STYLE_CODE CHAR(3)
MODIFIER_CODE CHAR(1)
DESCRIPTION VARCHAR(255) NN
PRICE_MULTIPLIER DECIMAL(3,55) NN
DISPLAY_ORDER INT NN
IS_ACTIVE CHAR(1) NN
CREATED_AT TIMESTAMP NN
(STYLE_CODE, MODIFIER_CODE)
CART_ITEMS
ID BIGINT
USER_ID BIGINT NN
INVENTORY_ID BIGINT NN
QUANTITY INT NN
UNIT_PRICE_CENTS INT NN
CURRENCY_CODE CHAR(3) NN
CREATED_AT TIMESTAMP NN
UPDATED_AT TIMESTAMP NN
INVENTORY
ID BIGINT
SERIES_ID BIGINT NN
SERIES_CODE CHAR(2) NN
STYLE_CODE CHAR(3) NN
SERIAL_NUMBER CHAR(4) NN
MODIFIER_CODE CHAR(3) NN
QUANTITY_AVAILABLE INT NN
UNIT_PRICE_CENTS INT NN
CURRENCY_CODE CHAR(3) NN
CREATED_AT TIMESTAMP NN
UPDATED_AT TIMESTAMP NN
(SERIES_CODE, STYLE, STYLE, SERIAL, NUMBER, MODIFIER_CODE)
ORDER_ITEMS
ID BIGINT
ORDER_ID BIGINT NN
INVENTORY_ID BIGINT NN
SELLER_ID BIGINT NN
SKU VARCHAR(255) NN
PRODUCT_NAME VARCHAR(255) NN
UNIT_PRICE_CENTS INT NN
CURRENCY_CODE CHAR(1) NN
QUANTITY INT NN
TOTAL_CENTS INT NN
CREATED_AT TIMESTAMP NN
CARTS
ID BIGINT
USER_ID BIGINT
STATUS VARCHAR(50) NN
CURRENCY_CODE CHAR(3) NN
CREATED_AT TIMESTAMP NN
UPDATED_AT TIMESTAMP NN
FK ADDRESSES_USERS
USERS
ID BIGINT
EMAIL VARCHAR(320) NN
HASHED_PASSWORD VARCHAR(320) NN
PASSWORD_ALG VARCHAR(50)
FIRST_NAME VARCHAR(64) NN
LAST_NAME VARCHAR(64) NN
IS_ACTIVE CHAR(1) NN
CREATED_AT TIMESTAMP NN
CREATED_AT TIMESTAMP NN
FK ADDRESSES_USERS
FK ADDRESSES_USERS
FK ORDER_ADDRESSES
ORDERS
ID BIGINT
USER_ID BIGINT
STATUS VARCHAR(23) NN
CURRENCY_CODE CHAR(1) NN
SUBTOTAL_COUNT INT NN
TAX_CENTS INT NN
DISCOUNT_CENTS INT NN
DISCOUNT_CENTS INT NN
TOTAL_CENTS INT NN
BILLING_ADDRESS_ID BIGINT
BILLING_ADDRESS_ID BIGINT
PLACED_AT TIMESTAMP
CREATED_AT TIMESTAMP NN
UPDATED_AT TIMESTAMP NN
ADDRESSES
ID BIGINT
USER_ID BIGINT
FULL_NAME VARCHAR(54) NN
LINE1 VARCHAR(54) NN
LINE2 VARCHAR(55)
CITY VARCHAR(54) NN
REGION VARCHAR(52)
POSTAL_CODE VARCHAR(54) NN
COUNTRY_CODE CHAR(3) NN
PHONE VARCHAR(44)
CREATED_AT TIMESTAMP NN
TOKENS
ID BIGINT
USER_ID DECIMAL NN
ORDER_ADDRESSES
ID BIGINT
FULL_NAME VARCHAR(M) NN
LINE1 VARCHAR(M) NN
LINE2 VARCHAR(32)
CITY VARCHAR(64) NN
REGION VARCHAR(58)
POSTAL_CODE VARCHAR(24) NN
COUNTRY_CODE CHAR(3) NN
PHONE VARCHAR(57)
CREATED_AT TIMESTAMP NN
FK ORDER_ORDER

### ECOMAPP database components

The ECOMDB01 database is organized into 15 tablespaces, with each table placed in its own space to support flexibility and management. It includes the USERS, TOKENS, SERIES, STYLES, STYLE_MODIFIERS, PRODUCT_TYPES, PRODUCT_VARIANTS, VARIANT_IMAGES, INVENTORY, CARTS, CART_ITEMS, ADDRESSES, ORDERS, ORDER_ADDRESSES, and ORDER_ITEMS tables, covering the core data structures of the system. Indexing is handled through primary key indexes along with business key indexes to maintain efficient and reliable query performance.

### Normalization

ECOMAPP NXTCG demonstrates a high level of normalization, with the schema aligning with Third Normal Form (3NF). All relations satisfy the requirements of First and Second Normal Forms, and the use of composite-key tables ensures that each non-key attribute is fully dependent on the entire primary key.

## Installing the ECOMAPP database

ECOMAPP database is hosted on Db2 for z/OS platform. You can install the ECOMAPP database within TSO/ISPF.

Before installing ECOMAPP, you must have these requirements:

- Access: You must have access to TSO/ISPF.
- Db2: Version 13, subsystem ID `<Your SSID>`, Db2 software HLQ `<Your DB2HLQ>`, and a Db2 Connect license.
- User Privileges: SYSADM/SYSCTRL or CREATEDB.
- Storage: 100 tracks for DDL library, 50 tracks for JCL library, 500 MB DASD for tablespaces and indexes.

Below are steps instructing you on how to install ECOMAPP database within TSO/ISPF.

1. Before uploading any files, you must customize the following placeholders throughout all DDL and JCL files:

| Placeholder | Description |
|---|---|
| **<Your HLQ>** | Your z/OS High-Level Qualifier |
| **<Your USERGRP>** | Application user group for DML access |
| **<Your SSID>** | Db2 subsystem ID |
| **<Your DB2HLQ>** | Db2 Software Library |

2. Create an Initial Partitioned Data Set (PDS) in `<Your HLQ>.ECOMAPP.ALOC`. You must create one dataset and upload the `ECOMALOC.jcl` as a member of that dataset.Your new dataset can be accessed by `<Your HLQ>.ECOMAPP.ALLOC(ECOMALOC)`. Configure your new dataset with the appropriate quantities, blocks, and record formattings below.

```
Command ===> A

Dataset Name: <Your HLQ>.ECOMAPP.ECOMALOC

Management class . . . <blank>

Storage class  . . . . <blank>

Volume serial  . . . . <blank>

Device type  . . . . . <blank or SYSDA>

Data class . . . . . . <blank>

Space units  . . . . . TRKS

Primary quantity . . . 5

Secondary quantity . . 2

Directory blocks . . . 2

Record format  . . . . FB

Record length  . . . . 80

Block size . . . . . . 3120

Data set name type . . PDS
```

   a. Log into TSO/ISPF and allocate the Initial PDS.
   b. Upload ECOMALOC.jcl via Zowe or by pasting in the code in ISPF. You can find and edit it by using option 3.4 to find `<Your HLQ>.ECOMAPP.ALLOC(ECOMALLOC)`.
3. Run ECOMALOC to create DDL and JCL libraries. This job creates the two main PDS libraries for the DDL and JCL files.
   a. Insert option 3.4 in DSLIST.
   b. Navigate to `<Your HLQ>.ECOMAPP.ALLOC(ECOMALOC)`.
   c. Type `SUB` next to the ECOMALOC member.
   d. Click Enter.
4. Upload all SQL DDL files to the `<Your HLQ>.ECOMAPP.DDL` library. You can use Zowe to upload members directly from the project repository into the dataset.
   a. Insert option 3.4 in DSLIST.
   b. Navigate to `<Your HLQ>.ECOMAPP.DDL`.
   c. Click Enter.
   d. Navigate to the ISPF main menu and insert Option 2.
   e. Insert `<Your HLQ>.ECOMAPP.DDL(<membername>)`.
   f. Paste in the new member contents.
5. Upload all JCL job files to the `<Your HLQ>.ECOMAPP.JCL` library using TSO/ISPF.

   a. Insert option 3.4 in DSLIST.
   b. Navigate to `<Your HLQ>.ECOMAPP.JCL`.
   c. Click Enter.
   d. Navigate to the ISPF main menu and insert Option 2.
   e. Insert `<Your HLQ>.ECOMAPP.DDL(<membername>)`.
   f. Paste in the new member contents.
6. Submit the ECOMDDL job to create the database. Submit an ECOMDDL Job in TSO/ISPF.
   a. Insert option 3.4 in DSLIST.
   b. Navigate to `<Your HLQ>.ECOMAPP.JCL`.
   c. Insert `B` for browsing to see the members.
   d. Type `SUB` next to the ECOMDDL member.
   e. Click Enter
7. Verify the database creation in TSO/ISPF.

   a. Insert option 2 in SPUFI.

   b. Enter `SELECT * FROM SYSIBM.SYSTABLES WHERE CREATOR = '<Your HLQ>';`.
      The result will show all of the tables in the database.

8. Initialize the Product catalog by submitting an ECPMINIT Job in TSO/ISPF.
   a. Navigate to `<Your HLQ>.ECOMAPP.JCL`.
   b. Type `SUB`next to ECOMINIT member.
9. Run statistics for optimal query performance by running ECOMRS.
   a. Navigate to `<Your HLQ>.ECOMAPP.JCL`.
   b. Type `SUB` next to ECOMRS member.
   c. Check the job output in SDSF job status menu.

# Adding tables to the ECOMAPP Database

In the ECOMAPP database you can add new tables by using TSO/ISPF in the Db2 z/OS platform.

1. Update EO3TABLE.sql
   a. Add `CREATE TABLE` statement.
   b. Add `CREATE UNIQUE INDEX` for the primary key.
   c. Add foreign key constraints if applicable.
2. Update E04INDEX.sql
3. Add performance indexes for new table in E04INDEX.sql.
4. Update E05CMNT.sql by adding a `COMMENT` statement for the new table and columns.

5. Add `COMMENT` statements for new tables and columns in E05CMNT.sql.
6. Update E06GRANT.sql by adding a `GRANT` statement for the new table.
7. Update E07LOAD.sql by adding an `INSERT` statement for the test data.
8. Update ECOMDROP.jcl by adding a `DROP TABLE` statement.
9. Re-upload and re-run.
10. Upload all the modified DDL files.
11. Submit ECOMDROP and ECOMDDL.

# Troubleshooting the NXTCG ECOMAPP database

There are common SQL and API error codes when configuring the ECOMAPP database. Solutions are provided for each issue.

### ECOMDDL Fails with RC=12 - PLAN NOT FOUND

**Cause:** Db2 plan DSNTEP3 not available or no EXECUTE privilege.

**Solution:**

```
1. Verify DSNTEP3 plan exists:

   -DIS THREAD(*) TYPE(PLAN)


2. Check your privileges:

   SELECT GRANTEE, GRANTEETYPE, AUTHHOWGOT

   FROM SYSIBM.SYSPLANAUTH

   WHERE NAME = 'DSNTEP3';


3. If missing privilege, have DBA grant:

   GRANT EXECUTE ON PLAN DSNTEP3 TO <Your HLQ>;
```

### ECOMDDL Fails with RC=12 - SQLCODE=-204 OBJECT NOT FOUND

**Cause:** DDL members not uploaded or incorrect dataset names.

**Solution:**

```
1. Verify DDL members exist:

   TSO/ISPF 3.4 → <Your HLQ>.ECOMAPP.DDL


2. Check member names match exactly:

   E01STGDB, E02TBSPC, E03TABLE, E04INDEX, E05CMNT, E06GRANT
```

```
   E07INIT1, E07INIT2, E07INIT3


3. Verify SET HLQ statement in ECOMDDL.jcl matches your dataset name
```

## ECOMDDL Fails with RC=8 - SQLCODE=-601 OBJECT ALREADY EXISTS

**Cause:** Database objects already exist.

**Solution:**

```
This is expected behavior when re-running ECOMDDL on existing database.

Job will continue and create any missing objects.


To start fresh:

1. Submit ECOMDROP first

2. Then submit ECOMDDL
```

## ECOMINIT Fails with RC=12 - Rest API failure with HTTP(S) status 500

**Cause:** SQL files have lines exceeding 72 characters.

**Solution:**

```
All E07INIT files in zOS_Datasets have been pre-processed to comply

with 72-character line limits. If you modified them:


1. Check for long lines:

   Edit member with ISPF

   COLS ON (show column numbers)

   Look for lines extending past column 72


2. Split long lines:

   - Break INSERT statements after table name

   - Continue column list on next line with leading spaces

   - Ensure VALUES clause on separate line


3. Re-upload corrected members
```

## ECOMINIT Fails with SQLCODE=-803 DUPLICATE KEY

**Cause:** Product catalog already loaded.

**Solution:**

```
This is expected behavior with RC=8. Job continues to next step.

Each step is conditioned on RC < 12.


To reload catalog from scratch:

1. Submit ECOMDROP first

2. Submit ECOMDDL

3. Submit ECOMINIT
```

## ECOMINIT Shows Wrong Row Counts

**Cause:** Previous data not cleared, or partial load.

**Solution:**

```
-- Check current countsSELECT'PRODUCT_VARIANTS'AS TABLE_NAME,COUNT(*)AS COUNT

FROM<Your HLQ>.PRODUCT_VARIANTS;


Expected: 4,824If count doesn't match:

1. Submit ECOMDROP

2. Submit ECOMDDL

3. Submit ECOMINIT
```

## NOT AUTHORIZED TO CREATE DATABASE

**Cause:** Insufficient Db2 privileges.

**Solution:**

```
Contact your Db2 DBA to grant CREATEDB authority:


GRANT CREATEDB ON SYSTEM TO <Your HLQ>;
```

## JOBLIB Libraries Not Found

**Cause:** Incorrect Db2 software library HLQ.

**Solution:**

```
1. Find correct Db2 HLQ on your system:

   TSO/ISPF 3.4 → DSN* (or DSNV*)

   Look for: SDSNLOAD, SDSNEXIT, RUNLIB.LOAD
```

```
2. Update JOBLIB statements in all JCL files:

   //JOBLIB   DD  DISP=SHR,DSN=<Correct HLQ>.SDSNEXIT

   //         DD  DISP=SHR,DSN=<Correct HLQ>.SDSNLOAD

   //         DD  DISP=SHR,DSN=<Correct HLQ>.RUNLIB.LOAD
```

## STORAGE GROUP NOT DEFINED

**Cause:** E01STGDB.sql uses default storage group.

### Solution:

```
1. Find available storage groups:

   SELECT * FROM SYSIBM.SYSSTOGROUP;


2. Update E01STGDB.sql:

   CREATE DATABASE ECOMDB01

       STOGROUP <Your STOGROUP>

       BUFFERPOOL BP0

       ...
```

## Price Calculations Return Wrong Values

**Cause:** Query not joining to STYLE_MODIFIERS table.

### Solution:

```
-- Incorrect (missing PRICE_MULTIPLIER):SELECT PT.BASE_PRICE_CENTS

FROM PRODUCT_VARIANTS PV

INNERJOIN PRODUCT_TYPES PT ON...;-- Correct (includes multiplier):SELECT

    PT.BASE_PRICE_CENTS,

    SM.PRICE_MULTIPLIER,

    CAST(PT.BASE_PRICE_CENTS * SM.PRICE_MULTIPLIER ASINT)AS FINAL_PRICE

FROM PRODUCT_VARIANTS PV

INNERJOIN PRODUCT_TYPES PT ON...INNERJOIN STYLE_MODIFIERS SM

    ON PV.STYLE_CODE = SM.STYLE_CODE

   AND PV.MODIFIER_CODE = SM.MODIFIER_CODE;
```

## Slow Queries

**Cause:** Statistics not gathered or outdated.

### Solution:

```
1. Submit ECOMRS to gather/refresh statistics

2. Verify statistics exist:

   SELECT NAME, CARD, NPAGES

   FROM SYSIBM.SYSTABLES

   WHERE CREATOR = '<Your HLQ>';


   If CARD = -1, statistics not gathered
```